

The Utility Problem

The practice of segmentation has been well-known in the distribution utility industry for many years. This type of segmentation usually includes consumer segmentation for consumer analytics. Consumer segmentation is usually run by the utility's Rates & Consumer Programmes department in order to improve customer programs and overall consumer satisfaction. The attributes which are used in such type of segmentation are attributes like tariffs, type of consumer, annual energy usage, type of heating, presence and type of behind-the-meter DER, etc. The resulting segments and elements (usually consumer or meters) assigned to these segments are often geographically distributed and don't share any other relationship except the commonality in previously mentioned attributes. While this type of segmentation is great for departments like Rates & Consumer Programmes, it is not sufficient for operations departments.

Operations departments such as Asset Management, Grid Planning or Revenue Protection approach the grid elements and consumers in a geographical-centered way. This approach might be by geographical location or further by grid topological relations. This approach is necessary due to the nature of activities which these departments are running in the distribution area. For example, the Grid Planning department is interested in certain feeders and loading conditions on these feeders. Feeder, in this case, is a set of elements which share the topological Geographical Information System (GIS) connectivity relation. Another example might be the Revenue Protection department, where the losses can be occurring downstream of certain MV/LV transformers. As the last example, the Standards department may want to deploy a power quality measurement device to a certain part of the grid (segment), which will be evaluated.

For these reasons, it is important to have a segmentation option which will divide the grid into topological segments, which are then manageable by operations departments. Such manageable segments then represent a set of grid locations (units) where the activities can be focused. The input and settings of the segmentation should allow for multiple options like a specification of the number of consumers per segment, number of transformers per segment, line length per segment, etc. These are all attributes which define what can and cannot be managed.

It is clear that the creation of the segmentation based on simple attributes such as tariff or type of consumer is not sufficient for a grid operations-focused department. A systematic approach to the creation of topological grid segmentation is required.

The Data Solutions and Methods

The Awesense Energy Transition Platform ingestion and data modelling of the GIS data is foundational for topological grid segmentation. The Awesense Platform allows dividing the grid into different-sized segments using the GIS connectivity and GIS elements attributes. By default, several options for the segmentation attributes are available in Awesense's True Grid

Intelligence (TGI) solution. These options can be set individually per segmentation or can be combined for each segmentation run. These options are:

- Number of sensors,
- Number of transformers,
- Sum of transformers kVA,
- Conductor/line length, etc.

Each of the above segmentation parameters can be assigned a weight for even more detailed segmentation options. All segment views are available directly in TGI on a map or in a table for further work with them. Detailed statistics for each created segment can also be displayed, with information such as the number of sensors, transformers, the sum of kVA, line lengths, etc. Everything is handled automatically, and the interface is very intuitive. No programming knowledge is required to create these segments, as everything can be defined very simply by selecting from a list of parameters and their values.

This functionality is made possible by the integration of geospatial data (GIS) that is part of the Awesense Energy Transition Platform. Resulted segments can be then used in multiple ways. First, the Segment Analysis can be applied to segmentation (TGI Segment Analysis) or the segmentation can be exposed through Awesense's Energy Data Model (EDM) APIs for further analytics using the preferred data analytical tool.

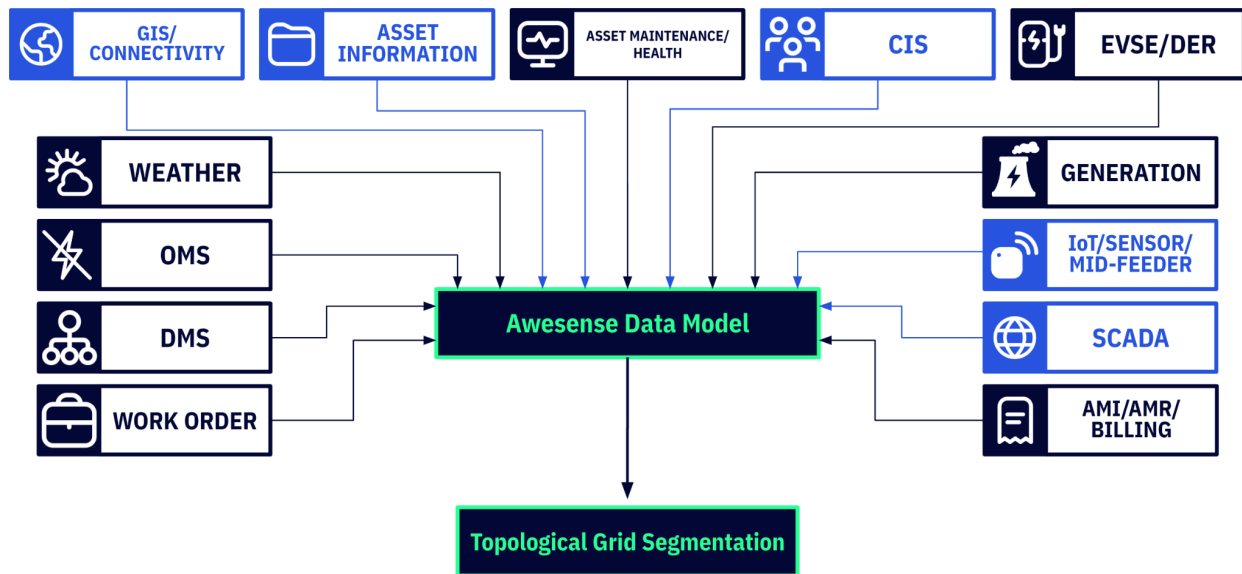
Topological Grid Segmentation

[UC22-01] - Awesense Energy Transition Platform Use Case



Required Data Sets

The required data for this use case are visualized in the following diagram. Data sources highlighted in [royal blue](#) are required for this use case.



Available Use Case Implementations:

Users of the Awesense Energy Transition Platform can access this use case by the following means:

1. TGI Segmentation - Please speak to Awesense about access to the sandbox environment.